

Your Presentation Title with a Subtitle Line

Your Name | February 25, 2026

Tsinghua University
Supervisor: Prof. Someone



Outline

- 1 Introduction & Motivation
- 2 Methodology
- 3 Experiments & Results
- 4 Case Study
- 5 Conclusion & Future Work

Introduction & Motivation

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Background

- Deep learning has achieved remarkable success across many domains¹
- The Transformer architecture revolutionised sequence modelling²
- Pre-trained language models further advanced NLP capabilities³

¹Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. “Deep Learning”. In: *Nature* 521.7553 (2015), pp. 436–444.

²Ashish Vaswani et al. “Attention Is All You Need”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2017, pp. 5998–6008.

³Jacob Devlin et al. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”. In: *arXiv preprint arXiv:1810.04805* (2019).

Research Question

Core Question

How can we improve X while maintaining Y under constraint Z ?

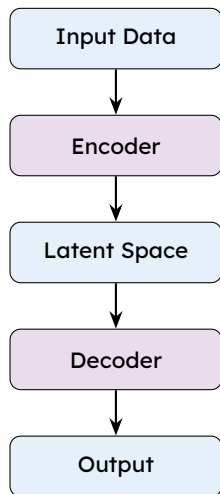
- **Challenge 1:** Existing methods overlook cross-layer interactions
- **Challenge 2:** Scalability remains an open issue for large models
- **Our approach:** A unified framework that addresses both challenges

Methodology

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Method Overview

- **Step 1:** Collect observations and build feature matrix X
- **Step 2:** Factorise $X \approx DR$ to discover latent patterns
- **Step 3:** Apply sparsity constraints for interpretability



Mathematical Formulation

Objective function

$$\mathcal{L}(D, R) = \underbrace{\|X - DR\|_F^2}_{\text{reconstruction}} + \lambda_1 \underbrace{\|R\|_1}_{\text{sparsity}} + \lambda_2 \underbrace{\Omega(D)}_{\text{regularisation}}$$

- $X \in \mathbb{R}^{m \times n}$: observation matrix (features $\times$ samples)
- $D \in \mathbb{R}^{m \times k}$: dictionary of k latent patterns
- $R \in \mathbb{R}^{k \times n}$: sparse representation coefficients
- λ_1, λ_2 : trade-off hyper-parameters

Algorithm: Alternating Optimisation

- 1 **Initialise** $D^{(0)}$ randomly; set $t = 0$
- 2 **Repeat** until convergence:
 - 1 Fix $D^{(t)}$, solve for $R^{(t+1)}$ via LASSO
 - 2 Fix $R^{(t+1)}$, update $D^{(t+1)}$ via projected gradient descent
 - 3 $t \leftarrow t + 1$
- 3 **Return** $D^{(t)}$, $R^{(t)}$

Complexity

Per-iteration cost: $\mathcal{O}(mnk)$. Typical convergence in 50–200 iterations.

Experiments & Results

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Experimental Setup

- **Models:** ModelA-Base, ModelB-Large
- **Datasets:** CIFAR-100, ImageNet-1K, GLUE benchmark
- **Hardware:** $4 \times$ NVIDIA A100 80GB
- **Training:** 100 epochs, AdamW optimiser, cosine LR schedule
- **Evaluation:** Top-1 accuracy, F1 score, inference latency

Main Results (Higher is Better)

Model	Method	AVG	Task A	Task B	Task C	Task D
ModelA	Full Model	0.892	0.921	0.873	0.901	0.872
	Baseline-1	0.826	0.854	0.812	0.835	0.805
	Baseline-2	0.838	0.861	0.825	0.842	0.823
	Ours	0.867	0.893	0.851	0.871	0.854
ModelB	Full Model	0.905	0.932	0.889	0.912	0.886
	Baseline-1	0.841	0.870	0.824	0.851	0.818
	Baseline-2	0.856	0.879	0.841	0.863	0.840
	Ours	0.881	0.908	0.867	0.885	0.865

- **Takeaway:** Our method consistently outperforms both baselines
- The gap is more pronounced on harder tasks (Task B, Task D)

Ablation Study

Variant	Accuracy	F1	Latency (ms)	Parameters (M)
Full method	0.881	0.874	12.3	125
w/o sparsity	0.862	0.854	12.1	125
w/o regularisation	0.871	0.863	12.4	125
Smaller dictionary	0.855	0.847	9.8	98
Larger dictionary	0.879	0.873	15.7	152

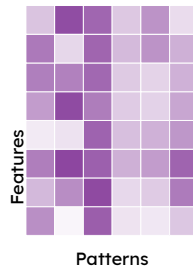
- Both sparsity and regularisation contribute to performance
- Dictionary size offers an accuracy-efficiency trade-off

Case Study

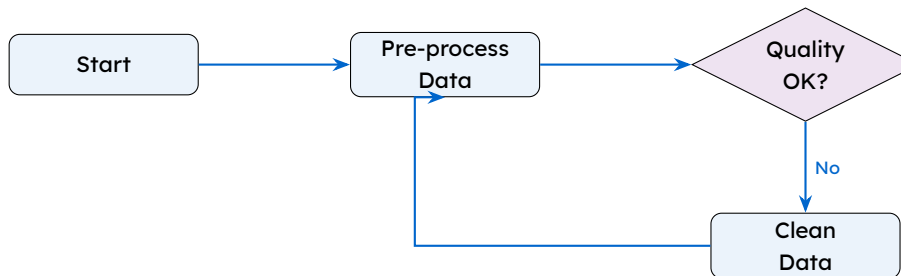
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Visualisation: Learned Patterns

- Each column of D corresponds to a latent pattern
- Semantically similar inputs activate overlapping patterns
- Patterns form a meaningful hierarchy: fine-grained $\rightarrow$ coarse-grained



TikZ Flowchart Example



Code Listing Example

Listing 1: Training loop (simplified)

```
1  import torch
2  import torch.nn as nn
3
4  model = MyModel(hidden_dim=256, n_layers=6)
5  optimizer =
        torch.optim.AdamW(model.parameters(),
        lr=3e-4)
6  criterion = nn.CrossEntropyLoss()
7
8  for epoch in range(100):
9      for batch in dataloader:
10         x, y = batch
11         logits = model(x)
12         loss = criterion(logits, y)
13         loss.backward()
```

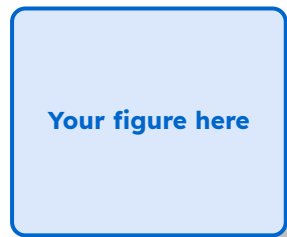


Using Columns & Images Together

- Left column for text explanations
- Right column for figures or diagrams
- Adjust column widths to taste

Tip

Use `\placeholder{text}` during drafting to reserve space for future figures.



Multi-Column Layout

Advantages

- Fast convergence
- Low memory footprint
- Easy to parallelise

Limitations

- Sensitive to initialisation
- Requires tuning λ
- Limited to linear decomposition

Summary

The method works best when the underlying structure is approximately low-rank and sparse.

Conclusion & Future Work

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Conclusion

- **Contribution 1:** We proposed a novel framework for discovering latent structures
- **Contribution 2:** Our approach achieves state-of-the-art results while remaining interpretable
- **Contribution 3:** Open-source tooling for interactive exploration

Future directions

- Extend to non-linear decomposition methods
- Scale to billion-parameter models
- Integrate with continual learning settings

Thank You!

Questions are welcome.

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GitHub: <https://github.com/your-username>

References I

- Devlin, Jacob et al. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”. In: *arXiv preprint arXiv:1810.04805* (2019).
- LeCun, Yann et al. “Deep Learning”. In: *Nature* 521.7553 (2015), pp. 436–444.
- Vaswani, Ashish et al. “Attention Is All You Need”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2017, pp. 5998–6008.